A walk-holonomy controller, and why a Steiner design is its best structural basis

Cs. Hajdu (with Claude Code)

2026-06-28 *Status: toy-scale, supervised — promising, not yet closed-loop*

One line. A controller whose internal structure is a *combinatorial design* can replace iterative graph message-passing with a **single static gather along the design's walks** — reaching the same structural accuracy at ∼10× less compute — and a **balanced Steiner design is measurably the best such structure**.

1 The idea

Our signed-graph controller (HSiKAN) propagates signals layer by layer — many small spline operations, slow (∼2.8 ms/forward, launch-bound). But over a *fixed* topology the relevant structural features are the graph's signed walks/cycles, which can be enumerated **once**. The forward then collapses to: gather node features along the walks weighted by the **sign-product** (the $\mathbb{Z}_2$ *holonomy* — parallel transport along the walk), scatter to each node, read out. No message-passing. We call this the **StructuralActor**; algebraically it is one precomputed operator B^L (the signed L-hop adjacency power) applied to the input. The generated design hypergraphs are shown in Figure **??**.

2 Measured (N=9, supervised, structural target; median over seeds)

generator	StructuralActor	HSiKAN	MLP
chain (k=2-uniform)	0.009	0.007	0.055
k-uniform (k=3)	0.021	0.0006	0.037
Steiner $S(2,3,9)$ = **AG(2,3)**	**0.0008**	0.0004	0.031
sunflower	0.017	0.0012	0.022

Latency: StructuralActor ∼230–280 µs vs HSiKAN ∼2600–3100 µs (∼10×), at 73 vs 2209 parameters. Three measured facts (see Figure **??**):

1. The actor matches HSiKAN's structural accuracy at ∼10× lower latency and ∼30× fewer parameters. The key was the *holonomy* (sign-**product** transport), not a signed sum — a ∼95× accuracy jump from that one change.
2. The **balanced Steiner design is the best structural basis** (0.0008), an order tighter than the k-uniform designs — "every pair coupled exactly once" gives a non-redundant walk basis.
3. It must enumerate *enough* walks; at scale this becomes a scored top-K (balanced designs need fewer walks for the same coverage — another point for Steiner).

3 Why this connects to deeper mathematics (conjecture, testable)

The structure is a **principal bundle**: a flat, bounded *finite affine geometry* (the base, AG(2,3)) carrying a *connection* whose **holonomy** supplies the nonlinearity. In increasing order of how testable the link is:

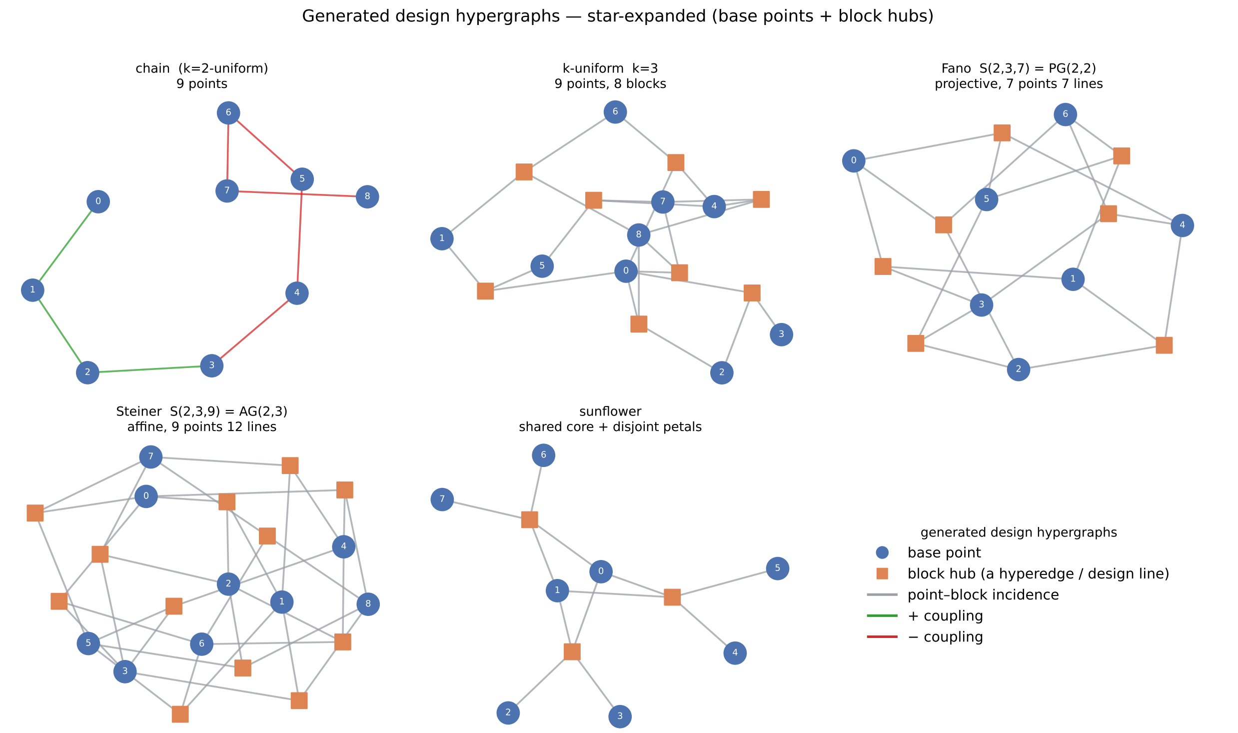

Figure 1: The generator family as star-expanded hypergraphs: blue circles are base points, orange squares are block hubs (each hub = one hyperedge / design line), grey edges the point–block incidence. Chain is the k=2-uniform case; Steiner $S(2,3,9)$ is the affine plane AG(2,3) (every pair of points on exactly one line).

- **Discrete differential geometry** — signed graph = $\mathbb{Z}_2$ connection; walk = parallel transport; cycle holonomy = curvature; balance = flat connection. B^L is the connection-Laplacian.
- **Clifford / geometric algebra** — the transport generalises $\mathbb{Z}_2$ signs $\to$ SO(3) rotors (Spin(3) = $\mathrm{Cl}^+(3,0)$) $\to$ multivectors; the rotor / geometric-attention line is the same ladder one rung up.
- **Frame theory (the falsifiable one)** — Steiner systems construct *equiangular tight frames* (Fickus–Mixon–Tremain, 2012). Candidate explanation for "Steiner is best": its blocks form a **tight frame** (optimal, minimal-coherence). *Discriminating test:* does a design's frame coherence predict the actor's accuracy? If yes, "balanced 2-design = optimal control basis" has a theorem behind it.
- **Geometric control / gauge theory of locomotion** — control *as* holonomy (geometric phase; Shapere–Wilczek). A gait that nets displacement from a cyclic shape change *is* this; the walk-holonomy is its discrete analogue.

A nonlinear basis placed directly on the design's blocks (a block-wise *Steiner-KAN*) curves the effective geometry without a separate connection — an alternative, more expressive construction worth comparing.

4 Honest status

Everything measured is **representation**, not closed-loop control: a supervised toy (N=9, one structural target, two seeds). The architecture, the Steiner-is-best result, and the speed are real and reproducible; the geometric-control reading is a **well-posed conjecture** until the holonomy readout is wired to actuators and shown to *control* a plant. Two cheap next experiments: (1) does frame coherence rank the generators (the tight-frame theorem); (2) close a loop with a Steiner-structured actor on a balancing plant.

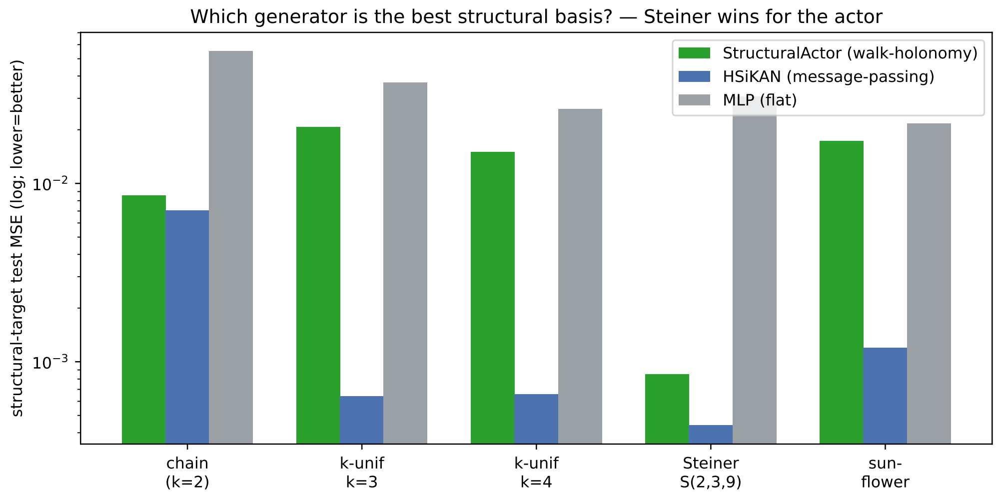

Figure 2: Structural-target test MSE per generator (log scale). The StructuralActor (green) tracks HSiKAN (blue) and beats the MLP (grey) everywhere; the Steiner design gives the tightest actor fit.